

CO₂ Assessment - 3

Sanjay Krishna R

192521055

Ethical Hacking - ITAHS

7/8/26

Scenario Based Assessment

Q.1

Footprinting Tools - Conducting Competitive Intelligence:

Info Required	Footprinting Tool
Domain registration design	WHOIS
Employee email address	the Harvester
Technology Stack	Wappalizer or Builtwith
Publicly exposed Subdomains	Amass or Sublist 3r

Tools Help in Competitive Intelligence

→ WHOIS provides ownership and registration info about the organization's domain.

→ the harvester gathers publicly available employee email addresses for organizational awareness.

→ Wappalizer / Builtwith reveals technologies and software used on public website.

Q.2

DNS Zone Transfer

Command

dig axfr @<DNS-Server-IP> example.com

Result:

- * If the server is secure, the transfer is denied.
- * If misconfigured, the DNS zone transfer succeeds and returns all DNS records.

Information obtained:

- A records - IP address of hosts
- MX Records - Mail Server Information
- NS Records - Name Server
- CNAME Records - Alias records.

Use During Reconnaissance:

- * Identifies internal and external hosts
- * Reveals mail server infrastructure
- * Discovers subdomains and network naming conventions.

Q.3

Social Engineering Attack and Countermeasures.

Phishing Attack:

The attacker impersonates the IT department and attempts to trick employees into entering their login credentials through a fake website.

Countermeasures:

- Verify the sender's email address carefully
- Do not click suspicious links or attachments
- check the website URL before entering credentials.
- Enable antivirus software and email filters.

Prevent credential theft:

- Verifying the sender helps detect fake emails.
- Confirming with IT prevents falling for impersonation.
- Reporting phishing emails protects other employees.